SECTION A (40 *marks*)

1. The probability that Mary speaks the truth is $^3/_5$ and that of Sarah is $^5/_8$. Find the probability that they will not agree on an identical point. (05 marks)

2. A car of mass 300kg moving at $144kmh^{-1}$ collides with a stationary trailer of mass 900kg thereby losing its momentum by 15%. Find the trailer's velocity after collision. (05 marks)

3. Use the trapezium rule with 6 ordinates to find the approximate value of $\int_0^{\pi/4} \frac{1}{1+\cos x}\,dx$, correct to three significant figures (05 marks)

4. The marks of 140 students are normally distributed with mean μ and variance 26. If 25 students scored below 42, find the value of μ. (05 marks)

5. A particle of mass 2kg resting on a rough horizontal plane is pulled by a force of magnitude $\frac{20\sqrt{3}}{3}$ N inclined at 60^0 to the horizontal. If the particle does not move, find the minimum value of the coefficient of friction between the particle and the plane.

6. Kamulasi has worked for 4 years and earns 400,000 shillings while Kakeeto has worked for 10 years in the same company and earns 800,000 shillings. Use linear inter polation / extra polation to estimate;
   i) the salary of Mukasa who has worked for 7 years in the same company. (03 marks)
   ii) the number of years Kakooza who earns one million shillings has worked for in that company. (02 marks)

7. The table below gives the grades of eight candidates in Mock and UNEB examinations 2024 in Mathematics paper 2.

Mock	E	C	B	F	D	A	B	O
UNEB	O	B	C	O	C	C	B	F

   i) Calculate the rank correlation coefficient for the data. (04 marks)
   ii) Comment on the significance at 1% level. (01 mark)

8. A particle with a mass of 2kg moves on a space curve such that at any time t its velocity is given by

   $V_{(t)} = (2 - 3t^2)i - 2\sin 2tj\ ms^{-1}$.

   Determine the power developed due to the force on the particle at any time t. (05 marks)

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SECTION B (60 *marks*)

*Attempt any **five** questions in this section.*

9. The table below shows the distribution of marks obtained in a certain subject.

Marks	Number of students
10 - <20	10
20 - <25	20
25 - <35	15
35 - <55	40
55 - <70	10
70 - <80	5

(a) Calculate the;

i) mean mark (03 marks)

ii) modal mark (03 marks)

(b) Draw a cumulative frequency curve and use it to estimate the;

i) Upper quartile of the marks.

ii) Pass mark if 60% of the students passed. (06 marks)

10. A particle travelling in a straight line with a constant acceleration covers distances S_1 and S_2 in the third and fourth seconds of its motion respectively.

(a) Show that its initial velocity is $\frac{1}{2}(7S_1 - 5S_2)$ (07 marks)

(b) If $S_1 = 2.5$m and $S_2 = 3.5$m, determine how far it goes in the 10th second of its motion. (05 marks)

11. (a) Real numbers A and B are approximated to a and b with maximum possible errors of E_1 and E_2 respectively. Show that the maximum percentage relative error made in approximating

$\frac{A^2}{B}$ by $\frac{a^2}{b}$ is $\left(2\left|\frac{E_1}{a}\right| + \left|\frac{E_2}{b}\right|\right) \times 100$ (06 marks)

(b) If a = 26.23 and b = 13.18 are calculated with percentage errors of 4 and 3 respectively. Determine the limits between which the exact value of $\frac{b}{b-a}$ is expected to lie, correct to 4 decimal places. (06 marks)

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12. (a) Events A and B are such that $P(^{A}/_{B}) = 0.2$, $p(^{A}/_{B'}) = 0.7$ and $P(B) = 0.6$.
Find $P(^{B'}/_{A'})$ (05 marks)

(b) Two tetrahedral dice both numbered 1 to 4 are thrown. If the die is un biased and the other is un balanced such that a four is twice likely to show up as any other number. Find the probability of obtaining a sum of five. (06 marks)

13. A brick of 2kg rests on a rough horizontal table (coefficient of friction 0.25). It is connected to another brock of mass 4kg which hangs freely in air by an inextensible string passing a smooth pulley at the edge of the table. If the system is released from rest;

(a) Represent the system on a force diagram. (02 marks)

(b) Determine the;

i) acceleration of the system (05 marks)

ii) tension in the string (02 marks)

iii) reaction of the pulley on the string (03 marks)

14. (a) By plotting a graph of x^3 and a suitable line on the same axes, show that the eq ıation $x^3 - 3x + 4 = 0$ has a real root between $^-2$ and $^-3$. (05 marks)

(b) Use the Newton Raphson method to find the root of the equation correct to 2 decimal places. (07 marks)

15. A continuous random variable T has a p.d.f given as

$$f(t) = \begin{cases} \frac{k}{9}t^2 & : 0 \leq t < 3 \\ k & 3 \leq t < 5 \\ 0 & \text{elsewhere} \end{cases}$$

(a) i) Determine the value of k. (03 marks)

ii) Sketch f(t) (02 marks)

(b) Determine the expression for $P(T \leq t)$. Hence find the 90th percentile. (07 marks)

16. (a) Forces of 10N, 15N and 25N, act along sides $\overrightarrow{AB}$, $\overrightarrow{BC}$ and $\overrightarrow{CA}$ of a triangle ABC in which $< BAC = 40^0$ and $< ACB = 50^0$. Determine the resultant force. (05 marks)

(b) Forces of magnitude 4p, 6p, p, 2p and 3p act in the direction $\overrightarrow{BA}$, $\overrightarrow{BC}$, $\overrightarrow{DC}$, $\overrightarrow{EF}$ and $\overrightarrow{FA}$ of a regular hexagon of side ametres.
Show that the forces form a couple. (07 marks)

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SECTION A: *(40 Marks)*

1. Show that $7_{C4} + 7_{C5} = 8_{C5}$ (5 marks)
2. Find the locus of the complex number $z = x + iy$. Which moves in the Argand diagram such that $\arg\left(\frac{z-3}{z-2i}\right) = \frac{\pi}{4}$ (5 marks)
3. Given that $y = \ln(x^2 - 5)$, show $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 2e^{-y}$ (5 marks)
4. Solve the differencial equation. $\frac{dy}{dx} = \left(\frac{x}{y}\right)e^{x+y}$ given that $y = 0$ when $x = 0$.(5 marks)
5. Find $\int \frac{1}{2x^2-12x+21}\,dx$ (5 marks)
6. The position vectors of points A and B are $3i-8j+k$ and $4k-2k$ respectively. Find the position vector of the foot of the perpendicular from the origin to AB. (5 marks)
7. Sketch the parabola given by the parametric equations $x=3t^2-2$, $y=6t+5$. Hence find its focus and directrix. (5 marks)
8. Solve the equation. $4\cot^2\theta - 3\operatorname{cosec}^2\theta = 2\cot\theta$ for $0^0 \leq \theta \leq 360^0$. (5 marks)

SECTION B: *(60 Marks)*

*Answer any **five** questions in this section.*

9. Prove that in triangle ABC

$$\frac{b-c}{b+c} = \frac{\sin B - \sin C}{\sin B + \sin C}$$

Deduce that $\tan \frac{1}{2}(B - C) = \frac{b-c}{b+c}\tan\frac{1}{2}(B + C) = \frac{b-c}{b+c}\cot\frac{1}{2}A$

Hence find B and C given that $A = 52\cancel{0}^0$, $b = 7$, $c = 5$. (12 marks)

10.(a) The sum of the first twenty terms of an arithmetic progression is 45 and the sum of the first forty terms is 290. Find the first term and common difference. (5 marks)

(b) If x is so small that its fourth and higher powers may be neglected. Show that $\sqrt[4]{1+x} + \sqrt[4]{1-x} = a - bx^2$ and find the numbers a and b, by putting $x = \frac{1}{16}$. Show that the sum of the fourth roots of 17 and 15 is 33.9985 approximately. (7 marks)

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11.(a) A savings scheme pays 5% per annum compound interest. A deposit of £100 is invested in this scheme at the start of each year.

i) Show that at the start of the third year and after the annual deposit has been made the amount in the scheme is £315.25. (2 marks)

ii) Find the amount in the scheme at the start of the fortieth year after the annual deposit has been made. (3 marks)

(b) The roots of the quadratic equation $x^2-px+q = 0$ are α and β. Form in terms of P and q, the quadratic equation whose roots are $\alpha^3 - P\alpha^2$ *and* $\beta^3 - P\beta^2$. (7 marks)

12.(a) Show that the equation of the plane through points A with position vector $-2i + 4k$ perpendicular to the vector $i + 3j - 2k$ is $x + 3y - 2z + 10 = 0$. (6 marks)

(b) i) Show that the vector $2i - 5j + 3.5k$ is perpendicular to the line

$$r = 2i - j + \lambda(4i + 3i + 2k)$$

ii) Calculate the angle between the vector $3i - 2j + k$ and the line in b(i) above. (4 marks)

13.The locus of P is such that the distance OP is half the distance PR where O is the origin and R is the point (-3, 6).

i) Show that the locus of P describes a circle in the x-y plane. (6 marks)

ii) Determine the radius and centre of the circle. (3 marks)

iii) Where does P cut the line $x = 3$? (3 marks)

14.(a) Find $\int \frac{x^2}{(x^4-1)}\,dx$ (7 marks)

(b) Evaluate $\int_0^1 \frac{x}{\sqrt{1+x}}\,dx$ (5 marks)

15. i) A curve is given parametrically by $x = \frac{1+1}{1-t}$ and $y = \frac{2t^2}{1-t}$.

Find the Cartesian equation of the curve. (3 marks)

ii) Sketch the curve. (6 marks)

iii) Find the area enclosed between the curve and the line $y = 1$. (3 marks)

16.(a) Integrate with respect to x.

$x^2\sin 2x$ (5 marks)

(b) The acceleration of a particle after time t seconds is given by $a = 5+\cos \frac{1}{2}t$. If initially the particle is moving at $1ms^{-1}$. Find the velocity after 2π seconds and the distance it would have covered by then.

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